LOSS ANALYSIS OF SUPERSTORE DATASET

A DATA-DRIVEN REPORT ON NET LOSS BY LOCATION, DATE, AND CATEGORY

Submitted by: Saad Thaplawala and Uzair Nadeem
Course/Instructor: Abeera Tariq

May 2025

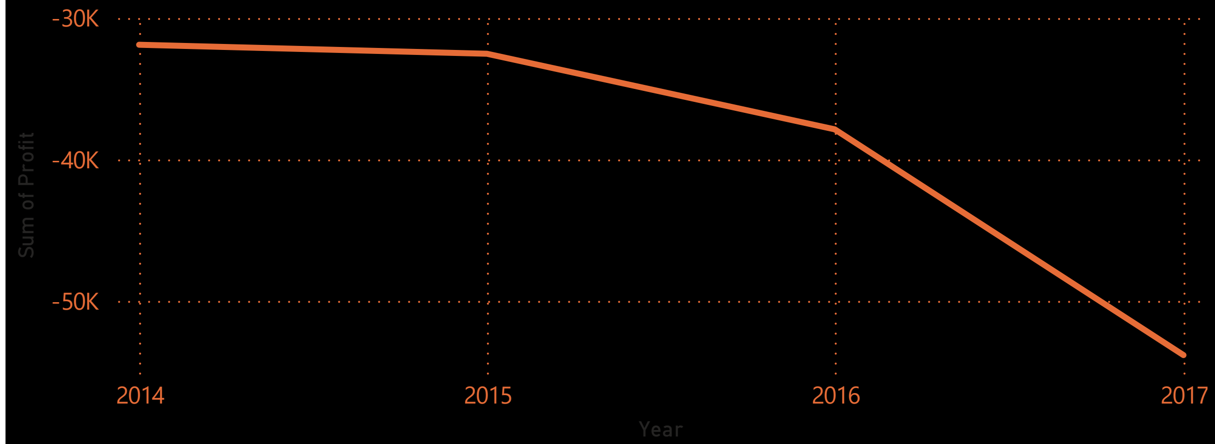
An Analysis of Loss Generating Locations and Products

The aim of this report is to analyze the profitability of the superstore across the worst performing locations and products. The analysis focuses on evaluating loss by:

- **Location:** Includes breakdown by Region, State, and City.
- **Category:** Includes Category and Sub-Category-level insights.
- **Date:** Time-based analysis using Order Date.

This analysis helps stakeholders understand which areas and product types are performing poorly, identify underperforming sectors, and optimize decision-making for logistics, marketing, and inventory management.

Total Loss by Year



Look Further into it

→

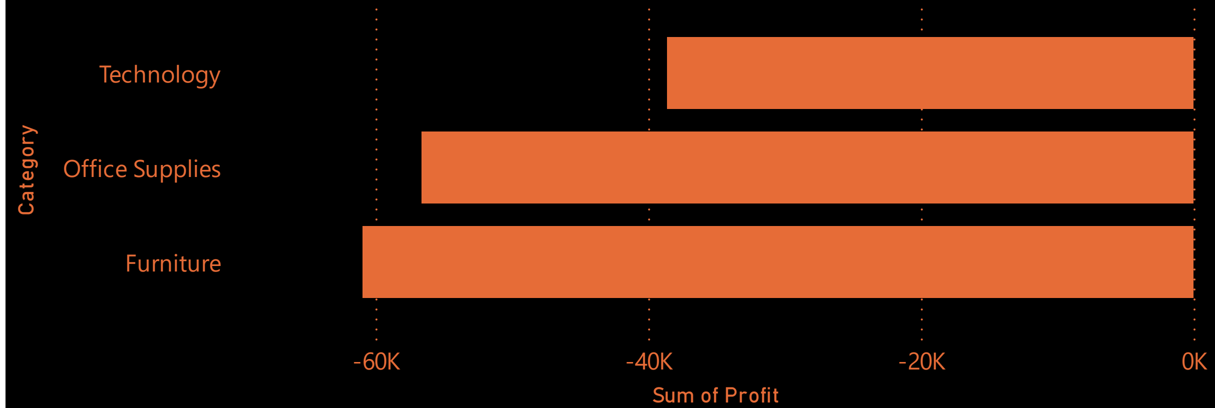
-156.13K

Sum of Profit

468.71K

Sum of Sales

Total Loss by Category



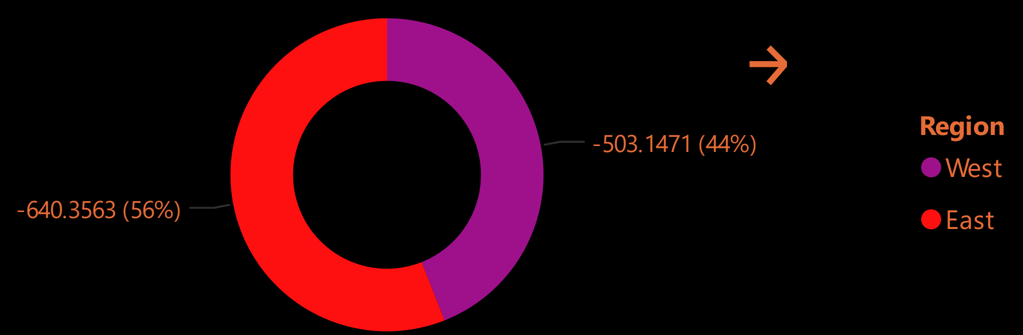
Look Further into it

→

Total Loss by Location

Look Further into it

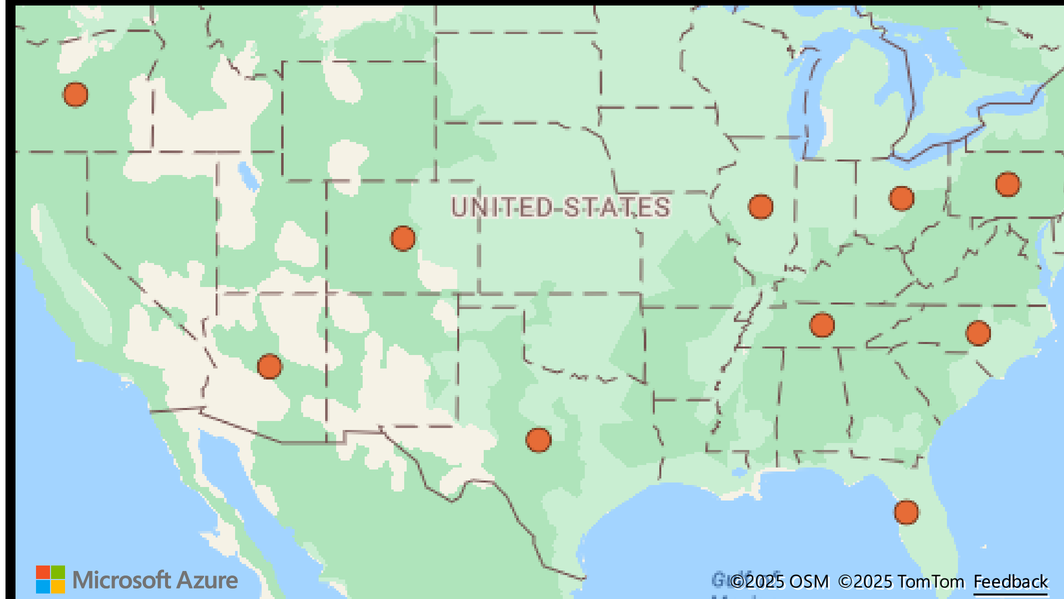
→



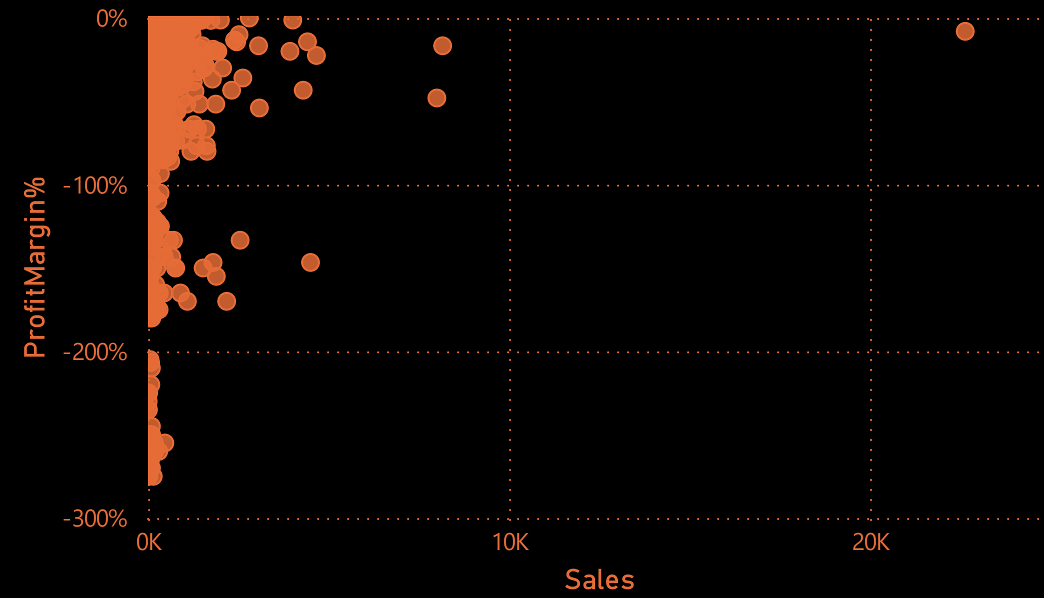
Key Insights:

- Overall, net profit has seen an increasing trend from 2014 - 2017, but so does total loss, increasing to its peak in 2017.
- The product category with the lowest net loss is Technology, and the category with the highest net loss is Furniture.
- The west and east regions contribute to total loss, with east being the major contributor.

Least Profitable States by Net Profit



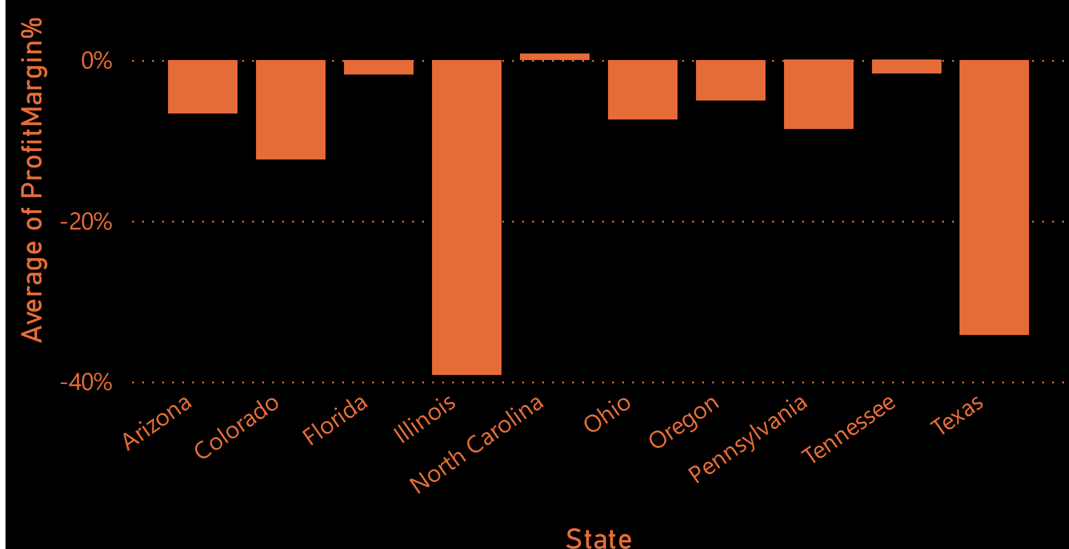
Sales and ProfitMargin%



Ship Mode

First Class	Second Class
Same Day	Standard Class

Average of ProfitMargin% by State (bottom 10)

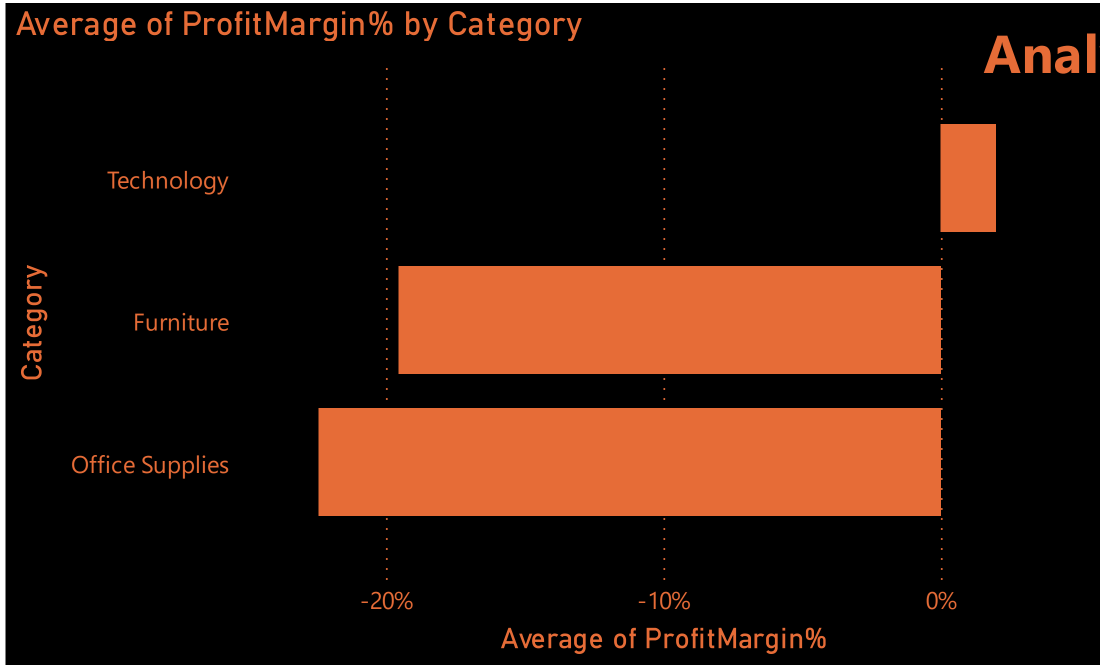


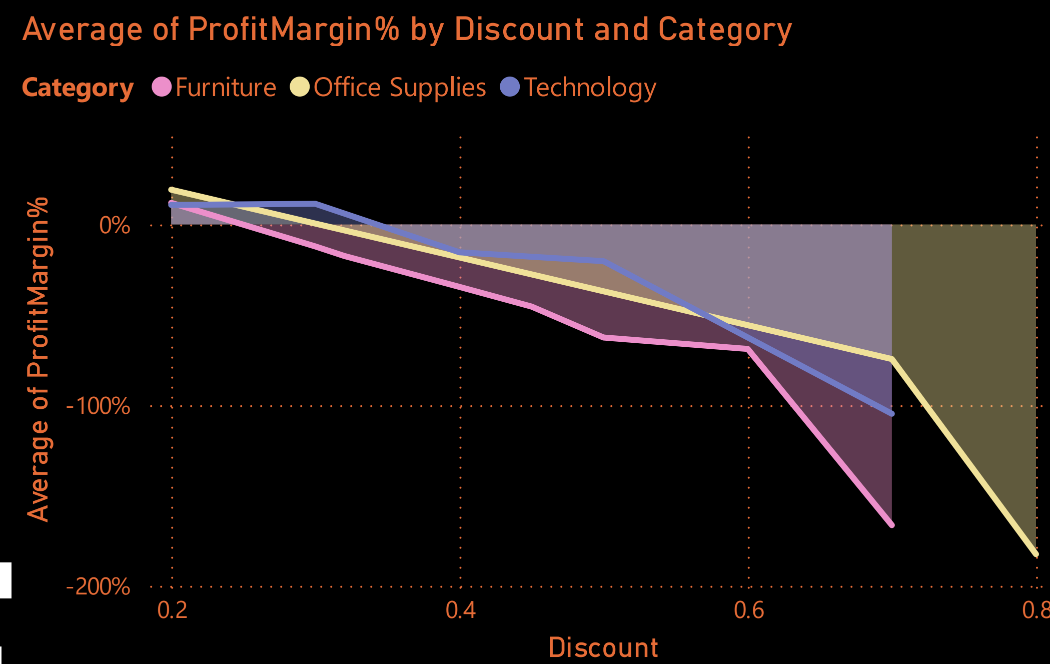
Segment

Consumer	Corporate	Home Office

- Texas, Ohio, Illinois, and Pennsylvania, are the top 4 loss generating states, and there are total of 10 states that incur a loss.
- Most sales in the lower bracket contribute to losses. Further breakdown can be seen on coming pages.
- Even though Texas has the highest loss overall, Illinois has the lowest percentage of profit margin(total profit to total sales).

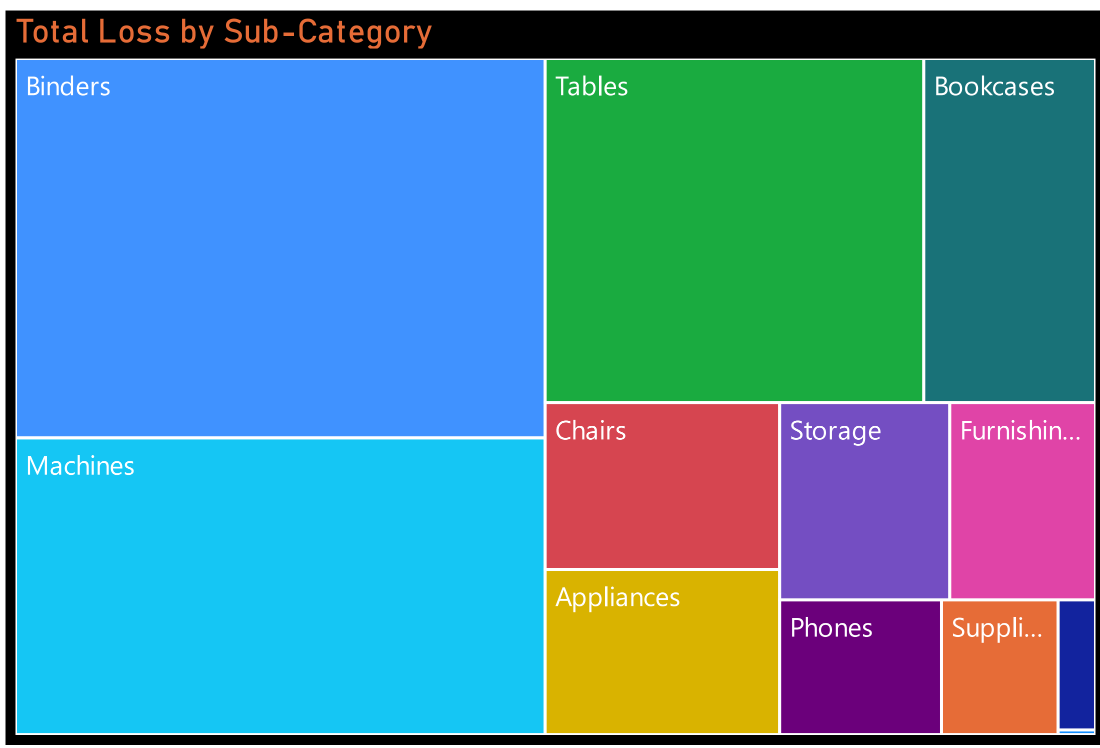
Analysis of Products Within Bottom 10 States on Prev. Page.





Ship Mode

First Class	Second Class
Same Day	Standard Class

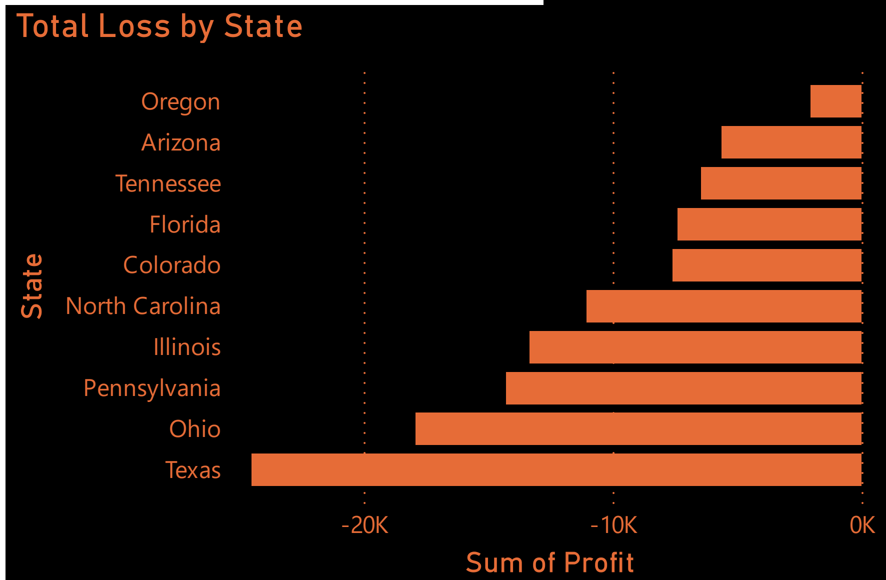


Segment

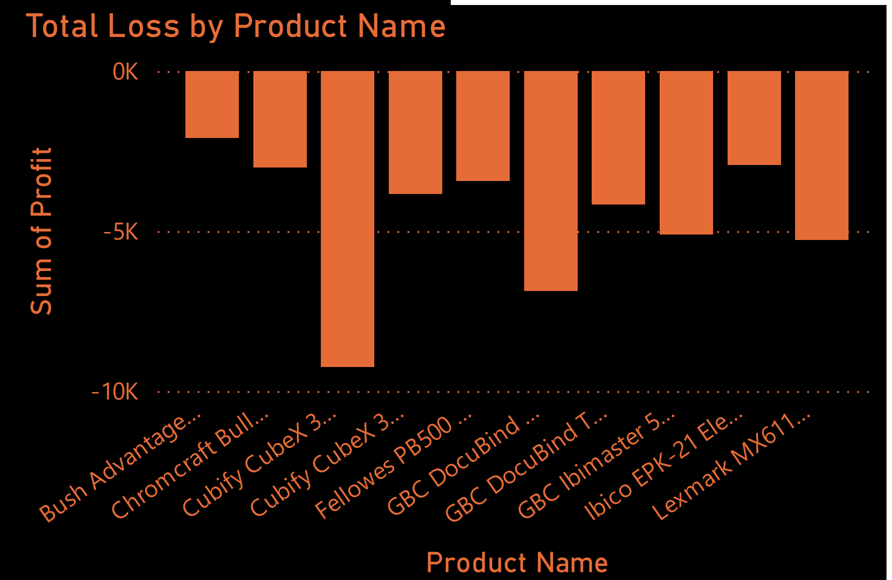
Consumer	Corporate	Home Office

- The bottom 5 sub-categories by total loss are binders, tables, machines, bookcases, and chairs.
- For highly discounted products, the profit margin is mostly negative, which means discounts contribute to losses.
- Overall, technology products have a positive profit margin, while furniture and office supplies have a significantly high percentage on the negative side.

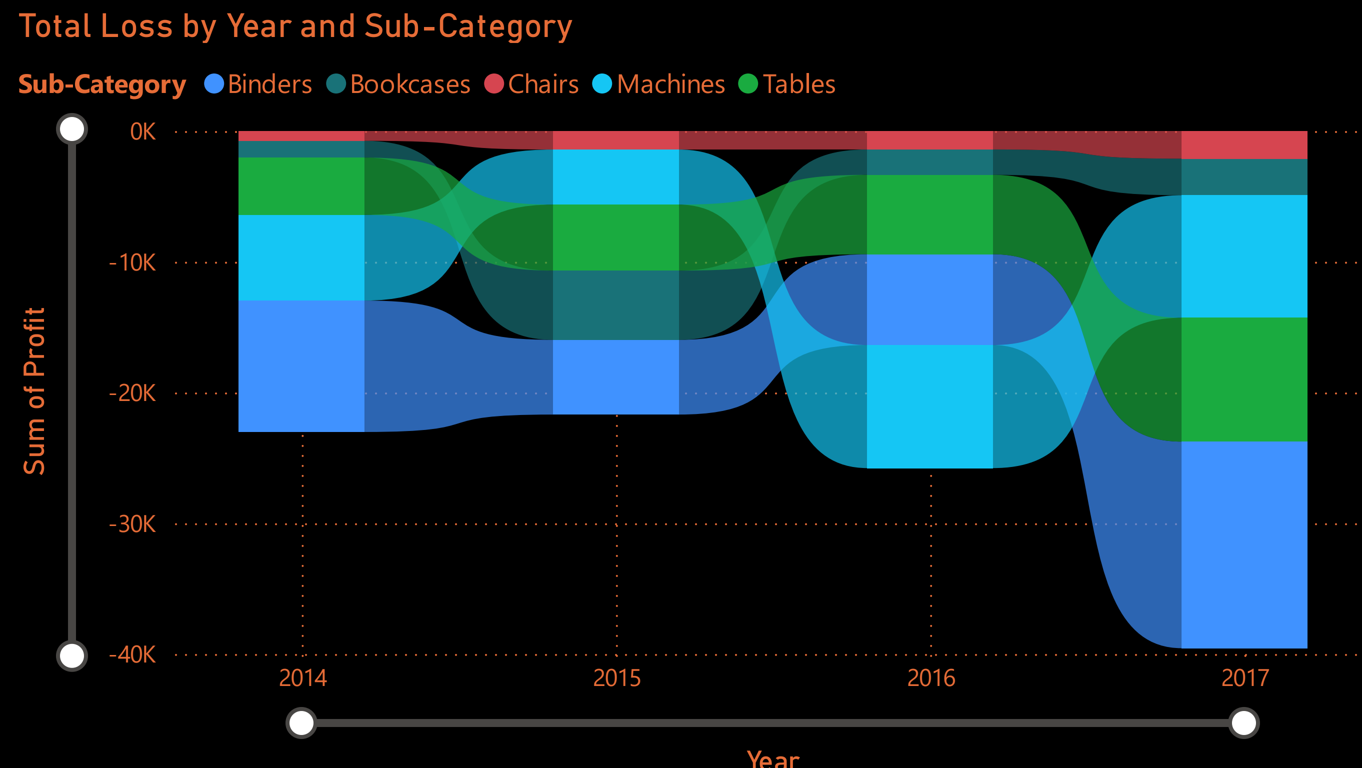
Analysis By Date of Categories and Products on Prev. Page



The products that have contributed the most to losses are Cubify CubeX 3D printers and GBC binding machines.



Binders have generated the highest loss across all years except 2016, with the highest loss incurred in 2017. While chairs have generated a loss in each year, it is significantly lower compared to other product sub-categories.



Click on any location or product in the charts above to view specific trends by date.

Move the sliders on the chart to the left to analyze data for specific dates or loss ranges.

Conclusions and Recommendations

1) Discount should be capped at a value that allows us to break even, rather than incur a loss.

2) Some states, especially Texas and Illinois have caused huge losses over the whole time frame with the product category causing the most issues being binders. The binders collection should either be evaluated to ensure product quality or be dropped from the inventory.

3)2017 is the year with the most overall profit but also the year with most loss, being around
$53000, which is a greater number than the profit generated in 2015.

Submission Date: 11 May 2025